Optimal Investment in Bitcoin Mining Farms: A Hedged Monte-Carlo Approach

Y. Kitapbayev* F. Macias† A. Alqubaisi‡ M. Lopez de Prado§
A. Lipton¶ J.P. Zubelli‖

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Abstract

The decision to invest in a Bitcoin mining facility involves navigating significant uncertainties, primarily driven by the high volatility of Bitcoin prices and fluctuating energy costs. This paper analyzes the optimal timing for such investments within a real options framework. We formulate the investment as an optimal stopping problem, considering both the option to defer the investment and the operational flexibility to temporarily suspend mining activities when unprofitable. To solve this complex valuation problem, we employ the Hedged Monte Carlo (HMC) method. HMC enhances the accuracy of the estimate by incorporating hedge sensitivities (Greeks) as control variables, thus reducing the variance of the Monte Carlo estimator. This approach allows for robust valuation under the real-world probability measure, accommodating complex dynamics for the underlying stochastic factors. We present a framework for determining the strategic value and optimal investment threshold for large-scale cryptocurrency mining projects

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HIGHLIGHTS

- Motivation: Evergroing Bitcoin cost due to energy consumption
- Determination of a suitable basis to tackle this problem (namely Margrabe type)
- Sustainability issues and energy consumptions
- Innovative usage of Real Options in the context of bit coin farming decisions
- Usage of the bitcoin index as a proxy to the actual bitcoin price.

IMPORTANT TOOL THE HASHPRICE `https://data.hashrateindex.com/network-data/bitcoin-hashprice-index`

Keywords: Real Options, Bitcoin Mining, Optimal Stopping, Hedged Monte Carlo, Investment under Uncertainty, Energy Finance.

*Affiliation and contact details.
†Affiliation.
‡Affiliation.
§Affiliation.
¶Affiliation.
‖Affiliation.

1. Introduction

The global expansion of Bitcoin mining operations represents a significant industrial development, characterized by substantial capital expenditures and high energy consumption. The economic viability of these ventures is inherently uncertain, being highly sensitive to the volatile market price of Bitcoin (BTC) and the cost of electricity. The scale of these investments is notable; for example, the establishment of a $ 2 billion crypto mining facility on Reem Island by the Phoenix group and the Abu Dhabi government underscores the strategic importance of robust valuation methods in this sector.

Given the irreversibility of the initial investment and the high uncertainty of future cash flows, traditional discounted cash flow (DCF) analysis is inadequate, as it fails to capture the value of managerial flexibility. Real Options Analysis (ROA) provides a superior framework, recognizing that investors possess critical flexibilities: the option to defer the investment (timing flexibility) and the option to suspend operations if the mining margin becomes negative (operational flexibility).

The main goal of this paper is to analyze the strategic value of investing in a Bitcoin mining farm under uncertainty. We model this decision as an optimal stopping problem with the objective of maximizing the expected net present value (NPV) by choosing the optimal time to execute the project.

Valuing these combined options (timing and switching) in the presence of multiple stochastic factors presents computational challenges. While traditional methods like binomial trees or finite difference methods exist, they often become intractable in higher dimensions or with complex dynamics. Monte Carlo simulation methods, particularly the Least-Squares Monte Carlo (LSM) approach [**?**], are frequently used, but can suffer from estimation biases and inefficiency.

To address these challenges, we utilize the Hedged Monte Carlo (HMC) method [**?** **?**]. HMC is a variance reduction technique that leverages information about the sensitivities (deltas) of the option value to the underlying assets. By incorporating the hedging strategy directly into the simulation as a control variate, HMC provides a more stable and accurate estimation of the continuation value and the optimal exercise boundary. Furthermore, HMC can be easily applied under the real-world probability measure $\mathbb{P}$, using a subjective discount rate, which avoids the complexities of estimating the market price of risk for non-tradable assets.

This paper contributes to the literature by applying the HMC methodology to the specific context of Bitcoin mining investment decisions, offering a robust and flexible tool for strategic planning in this emerging industry.

2. Literature Review

Fast mean-reverting stochastic volatility. Asymptotic analysis of invest-or-wait problems under fast mean-reverting stochastic volatility shows that stochastic volatility raises the value of waiting and shifts the optimal investment boundary relative to constant-volatility benchmarks [7].

Mean-reverting project value and investment cost. When both the project value and the investment cost are mean-reverting, the optimal trigger becomes genuinely two-dimensional; simple value-to-cost ratio rules hold only in special cases, and the problem is addressed numerically [4].

Hedged Monte Carlo in incomplete markets. For settings driven by historical/simulation scenarios—common in commodities—Zubelli and coauthors propose a hedged Monte Carlo approach that prices the real option jointly with a dynamically learned hedge [2].

Energy application: LNG contracts. These ideas extend to energy, where robust, risk-aware valuation and management of LNG contracts with cancellation options are formulated and solved within an optimization framework [3].

3. Problem Formulation

We formalize the investment decision faced by a company considering the establishment of a Bitcoin mining farm.

3.1. Investment Setup

Consider an investment horizon $[0, T]$. The company has the option to build a mining farm, requiring a fixed, irreversible sunk cost $K > 0$. If the company invests at time $\tau \in [0, T]$, the farm begins operations immediately and has a fixed operational lifetime T'. Consequently, the farm operates during the interval $[\tau, \tau + T']$.

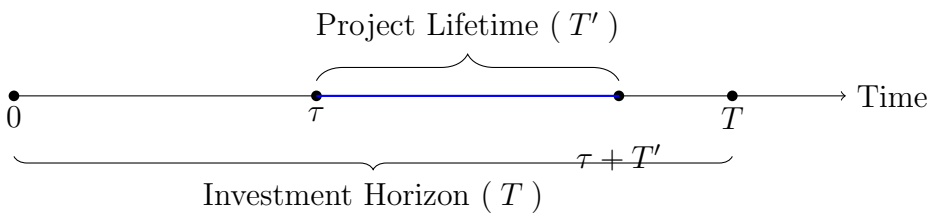

Figure 1: Timeline of the investment decision and project operation.

3.2. Stochastic Variables and Cash Flows

The profitability of the mining farm is driven by two primary stochastic processes:

- X_t : The price of 1 BTC in USD at time t.
- Y_t : The electricity price in USD/MWh at time t.

We assume the dynamics of X_t and Y_t are known or simulated under the real-world measure $\mathbb{P}$. Valuation is performed using a discount rate ρ, reflecting the company's cost of capital (WACC).

3.2.1. Net Present Value (NPV) of the Project

If the investment is made at time τ, the Net Present Value (NPV) of the project, evaluated at time τ, is given by:

$$G(x, y) = \mathbb{E}^{\mathbb{P}}_{x,y}\left[\int_{\tau}^{\tau+T'} e^{-\rho(t-\tau)}\left((X_t - \kappa Y_t)^+ - k\right) dt\right] - K \tag{3.1}$$

where $\mathbb{E}^{\mathbb{P}}_{x,y}[\cdot]$ denotes the expectation under $\mathbb{P}$ conditional on $X_\tau = x$ and $Y_\tau = y$. The parameters are:

- κ : The electricity consumption factor (MWh required to mine 1 BTC per unit time), capturing hardware efficiency and network difficulty.
- k : Fixed operating costs per unit time (e.g., maintenance, wages).
- K : The initial sunk cost.

We assume the firm has the operational flexibility to suspend mining (switch to an idle state) and resume operations without cost. The optimal strategy is to mine if $X_t > \kappa Y_t$, and stay idle otherwise. The instantaneous variable cash flow is thus $(X_t - \kappa Y_t)^+ := \max(X_t - \kappa Y_t, 0)$. The fixed cost k is incurred regardless of the operational state.

If the processes X_t and Y_t are time-homogeneous, the NPV function G depends only on the current values (x, y).

3.2.2. Monte Carlo Estimation of NPV

The NPV function $G(x, y)$ must generally be estimated using Monte Carlo simulation. Discretizing the lifetime $[0, T']$ into J intervals of length $h = T'/J$, the MC estimator is:

$$\widehat{G}(x, y) \approx \frac{h}{N} \sum_{n=1}^{N} \sum_{i=1}^{J} e^{-\rho t_i} \left(\left(X_{t_i}^{(n)} - \kappa Y_{t_i}^{(n)} \right)^+ - k \right) - K \tag{3.2}$$

where N is the number of simulated paths starting from (x, y). Note that evaluating $G(x, y)$ is computationally intensive, requiring a nested simulation whenever the option value is calculated.

4. The Hedged Monte Carlo Methodology

The objective is to find the optimal investment time τ to maximize the expected discounted value of the project. This is formulated as an optimal stopping problem:

$$V(x, y) = \operatorname{ess\,sup}_{0 \le \tau \le T} \mathbb{E}^{\mathbb{P}}_{x,y} \left[e^{-\rho\tau} G(X_\tau, Y_\tau) \right] \tag{4.1}$$

where the supremum is taken over all stopping times τ adapted to the filtration generated by (X_t, Y_t).

We apply the Hedged Monte Carlo (HMC) method to solve this problem. HMC utilizes dynamic programming and incorporates hedging information to improve the estimation of the continuation value.

4.1. Discretization and Backward Induction

We discretize the investment horizon $[0, T]$ into L intervals of length $\Delta = T/L$, with time points $t_j = j\Delta$. We simulate M independent trajectories of (X, Y) over this grid.

The algorithm proceeds via backward induction. At the terminal time $T = t_L$, the value is:

$$V(L, m) = G(X_T^{(m)}, Y_T^{(m)}) \tag{4.2}$$

for the m -th simulated path.

At time step t_j , the company compares the immediate investment value, $G(X_{t_j}, Y_{t_j})$, with the continuation value, $CV(j, X_{t_j}, Y_{t_j})$. We approximate the continuation value using a regression on a set of basis functions $\{C_a(x, y)\}_{a=1}^{B}$:

$$CV(j, x, y) \approx \sum_{a=1}^{B} \gamma_a^j C_a(x, y) \tag{4.3}$$

4.2. HMC Regression

In the HMC approach, the coefficients (γ_a^j) and the hedging coefficients (β_a^j, δ_a^j) are determined simultaneously by minimizing the squared replication error of the hedged portfolio. This minimization leverages the fact that the discounted option price process, when appropriately hedged, behaves like a martingale.

The objective function to minimize is:

$$\begin{aligned}\min_{\gamma,\beta,\delta} \sum_{m=1}^{M} \Bigg(& e^{-\rho\Delta} V\left(j+1, X_{t_{j+1}}^{(m)}, Y_{t_{j+1}}^{(m)}\right) - \sum_{a=1}^{B} \gamma_a^j C_a\left(X_{t_j}^{(m)}, Y_{t_j}^{(m)}\right) \\ & + \sum_{a=1}^{B} \beta_a^j \frac{\partial C_a}{\partial x}\left(X_{t_j}^{(m)}, Y_{t_j}^{(m)}\right)\left[X_{t_j}^{(m)} - e^{-\rho\Delta} X_{t_{j+1}}^{(m)}\right] \\ & + \sum_{a=1}^{B} \delta_a^j \frac{\partial C_a}{\partial y}\left(X_{t_j}^{(m)}, Y_{t_j}^{(m)}\right)\left[Y_{t_j}^{(m)} - e^{-\rho\Delta} Y_{t_{j+1}}^{(m)}\right]\Bigg)^2 \end{aligned} \tag{4.4}$$

The terms involving β_a^j (hedging X) and δ_a^j (hedging Y) act as control variates. They represent the P&L of delta-hedge portfolios constructed using the known sensitivities of the basis functions. By incorporating these "perfect hedges" for the basis functions, HMC significantly reduces the variance in the estimation of the continuation value.

4.3. Option Value Update

Once the coefficients $\hat{\gamma}_a^j$ are estimated, the continuation value is calculated. The project value at time t_j for path m is then:

$$V(j, m) = \max\left(G(X_{t_j}^{(m)}, Y_{t_j}^{(m)}), \sum_{a=1}^{B} \hat{\gamma}_a^j C_a\left(X_{t_j}^{(m)}, Y_{t_j}^{(m)}\right)\right) \tag{4.5}$$

This process is repeated backwards until t_0 , yielding the value of the investment option.

5. Numerical Implementation and Case Study

5.1. Strategic Baseline: The Hybrid Hosting Model

To evaluate the investment, we adopt a "Hybrid" or "Colocation" investment strategy, which represents the most common and practical entry point for mid-scale mining operations today. In this model, the investor retains ownership of the mining hardware (capital expenditure) but outsources the infrastructure and energy procurement (operational expenditure).

- **Capital Investment:** The firm purchases 350 modern, high-efficiency mining rigs (e.g., Antminer S19 XP, 140 TH/s) at an adjusted secondary-market price of $2,500/unit. This generates a total network computing power of approximately 49 PH/s. The initial sunk cost (K) for the hardware totals **$875,000**, avoiding the high interest premiums associated with full hardware leasing.

- **Operational Structure:** The hardware is deployed in a third-party hosting facility at an all-inclusive rate of $0.065/kWh. This rate removes the need for capital investment in transformers, land development, and maintenance staff. Consequently, the daily fixed operational cost (k) is assumed to be negligible ($k \approx 0$), as costs are fully variable based on energy consumption. The facility's total daily energy conversion factor (κ) is calculated as **25,284 kWh/day**.

Assuming current market conditions remain completely static (e.g., a deterministic Hashprice of approximately $45/PH/day), the project generates a positive monthly net profit of **$17,092**. This static scenario results in a traditional payback period of **4.3 years**.

5.2. Scenario 1: Deterministic Project Value and Optimal Exit

Before introducing stochastic volatility, we establish a strict deterministic floor by calculating the project's optimal static exit timing. This analysis evaluates the Net Present Value (NPV) of the investment assuming a variable stopping time T . The total NPV incorporates both the discounted operational cash flows accumulated up to month T and the salvage (resale) value of the mining hardware at that time. Given the rapid technological obsolescence inherent to ASIC miners, we assume the hardware depreciates at an aggressive rate of **4%** per month.

Assuming constant market conditions (zero volatility in both Hashprice and energy costs), the project's value profile is concave. This reflects the fundamental trade-off between accumulating steady operational profits and the continuous degradation of the hardware's resale value.

As depicted in Figure 2, the deterministic project value is maximized if the investor plans to liquidate the operation and sell the hardware at **Month 60**. This strategy yields a maximum theoretical NPV of **$89,586**. This figure establishes the baseline value of the project, completely ignoring the strategic value of reacting to crypto-market volatility.

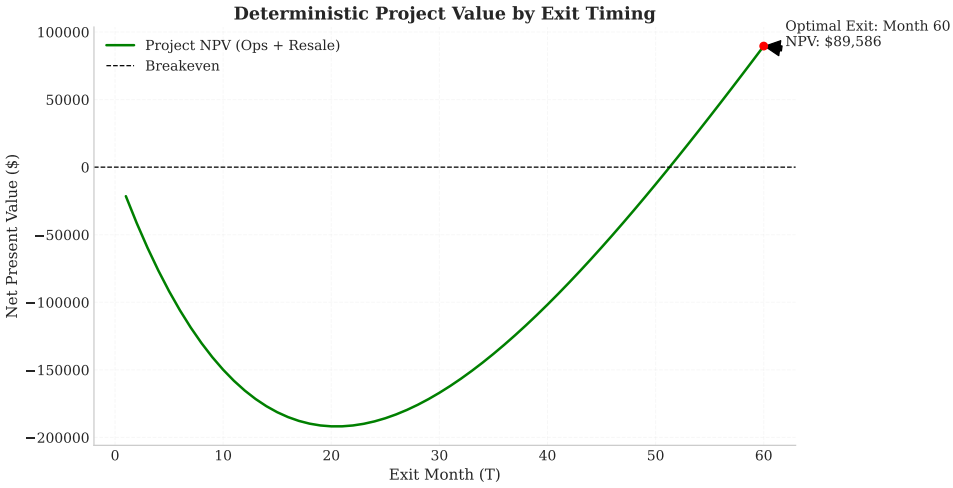

Figure 2: Deterministic Project NPV as a function of the stopping time T. The curve illustrates the trade-off between accumulating operational cash flows and the monthly depreciation of the hardware's resale value.

6. Results and Discussion

6.1. Scenario 2: The Energy Constant Case (Stochastic Revenue)

In our first stochastic scenario, we isolate the value of *Revenue Volatility*. Because the firm pays a fixed electricity hosting rate ($0.065/kWh), energy price risk is eliminated. However, the project remains fully exposed to the extreme volatility of Bitcoin prices (X_t) and the resulting Hashprice (Z_t).

Historical Calibration and Simulation To accurately model revenue uncertainty, we first calibrate our Geometric Brownian Motion (GBM) parameters using historical market data. Figure 3 illustrates the historical co-movement of Bitcoin prices and the Hashprice index. The strong empirical correlation justifies modeling them as coupled stochastic processes within our framework.

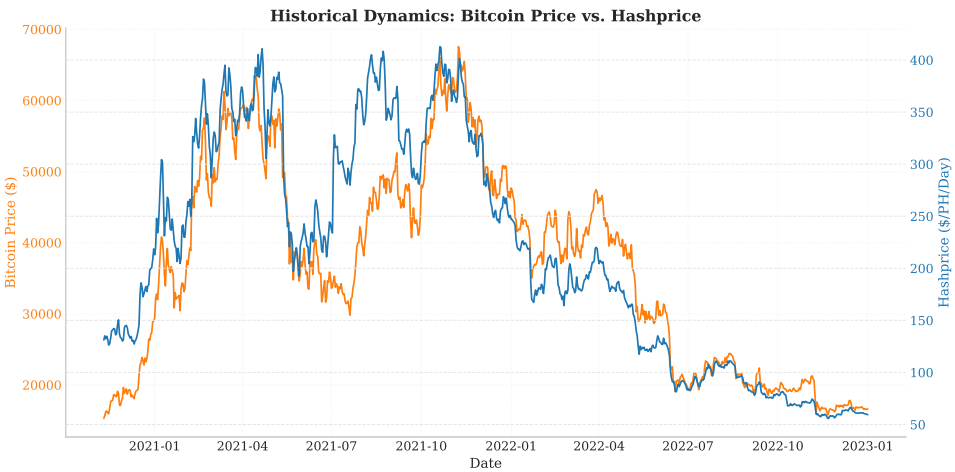

Figure 3: Historical dynamics of Bitcoin Price and Hashprice. The tight coupling of these metrics serves as the calibration basis for the covariance matrix.

Using these calibrated parameters, we project thousands of potential future paths over the project's lifetime. Figure 4 displays the resulting "cone of uncertainty" for both variables. The wideness of these confidence intervals visually underscores the significant risk of a "crypto winter" (where revenues drop below the fixed hosting costs) as well as the potential for massive upside.

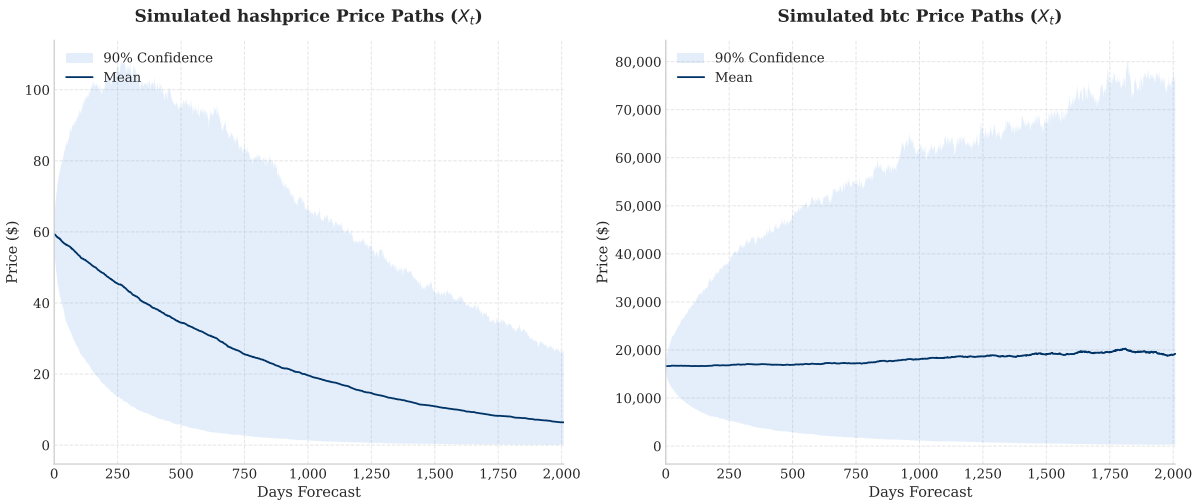

Figure 4: Simulated Monte Carlo paths for Bitcoin (X_t) and Hashprice (Z_t), displaying the 5% and 95% confidence intervals over the forecast horizon.

Real Options Valuation and The Switching Option Crucially, the firm possesses an operational *Switching Option*: management can temporarily suspend mining operations if the Hashprice drops below the hosting break-even threshold ($Z_t < \kappa Y_{fixed}$). This prevents negative cash flows, allowing the firm to safely hold the hardware and resume operations when profitability returns.

We evaluate this flexibility using the Hedged Monte Carlo (HMC) algorithm, which calculates the optimal early-exercise (suspension) boundary dynamically across the simulated market paths. Figure 5 demonstrates the divergence between the total option value and the intrinsic value over time.

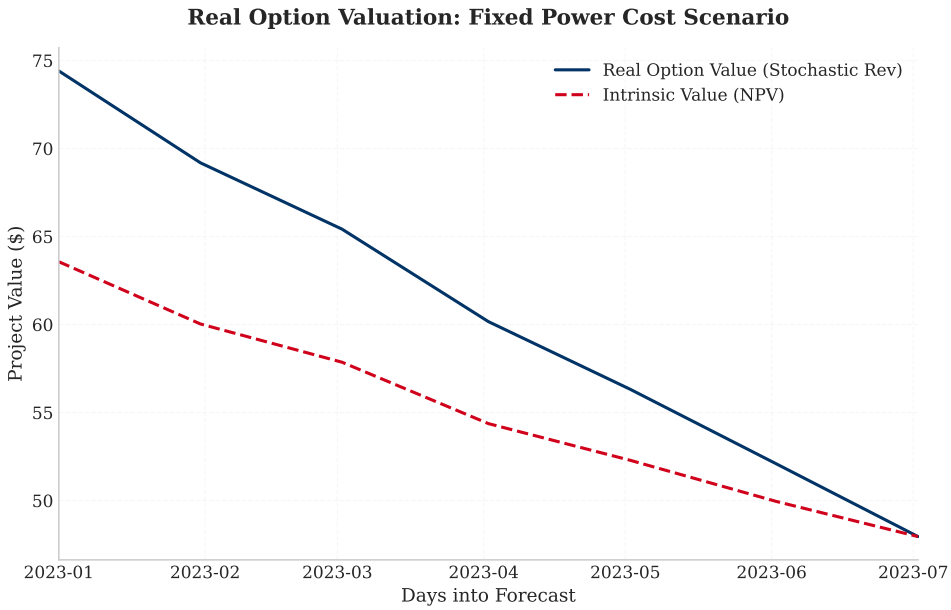

Figure 5: Evolution of the Real Option Value versus Intrinsic Value (Traditional NPV) over the projected timeframe for the Fixed Cost scenario.

When evaluated purely intrinsically—forcing immediate exercise without the flexibility to wait or switch—the traditional NPV under simulated paths is estimated at **$64**. However, when properly valuing the operational flexibility to weather market downturns and capitalize on bull runs, the Real Option Value is calculated at **$74**.

The difference between these two metrics yields a flexibility premium of **$11**. This significant premium confirms that traditional NPV severely underprices mining assets by ignoring management's ability to truncate downside risk through operational suspension.

6.2. Scenario 3: The "Grid" Case (Full Stochasticity)

In the final scenario, the firm bypasses the fixed-rate hosting provider and connects directly to the wholesale electricity grid (e.g., ERCOT in Texas). This exposes the operation to significant energy price volatility (Y_t). However, it also grants the firm the *Spark Spread Option*—the ability to dynamically curtail operations during acute power price spikes while benefiting from significantly lower average off-peak rates.

Energy Volatility and The Hedging Proxy Unlike Bitcoin, electricity cannot be efficiently stored, resulting in spot prices that exhibit extreme jumps and seasonality. To value the option to curtail power, the Hedged Monte Carlo (HMC) engine requires a tradable asset to construct a replicating portfolio. Because short-term electricity futures are often illiquid, we utilize historical Natural Gas futures as a highly correlated proxy for ERCOT spot prices.

Figure 6 illustrates the historical volatility of these energy costs. By calibrating the covariance matrix between Bitcoin returns and this energy proxy, the algorithm learns how to dynamically delta-hedge the dual risk.

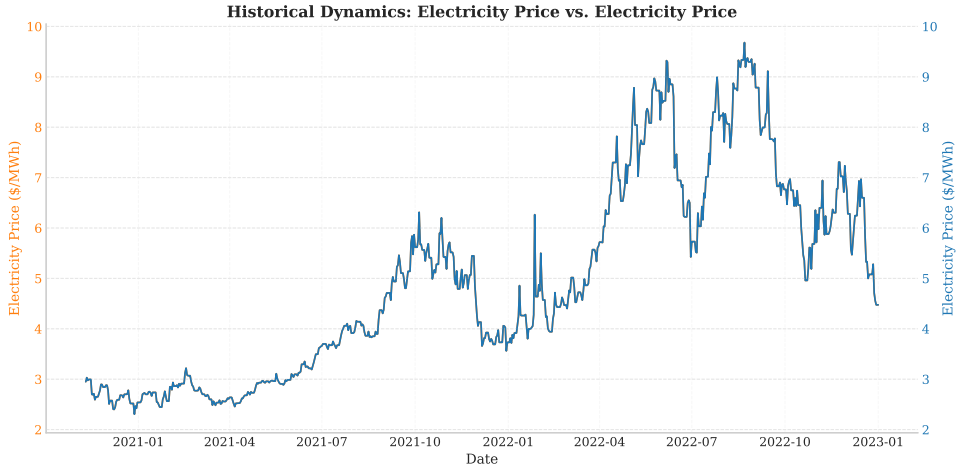

Figure 6: Historical dynamics of wholesale electricity proxy prices. The extreme right-tail skewness (price spikes) highlights the necessity of the curtailment option.

Real Options Valuation and The Spark Spread In this dual-stochastic environment, the optimal stopping boundary becomes two-dimensional. The algorithm evaluates whether the immediate mining revenue exceeds the spot electricity cost ($X_t > \kappa Y_t$). During severe energy price spikes (common during summer months or extreme winter storms), the model triggers a temporary shutdown, avoiding massive operational losses.

We simulate thousands of joint paths for X_t , Z_t , and Y_t to calculate the final project value. As shown in Figure 7, embracing energy volatility significantly alters the strategic value profile of the project.

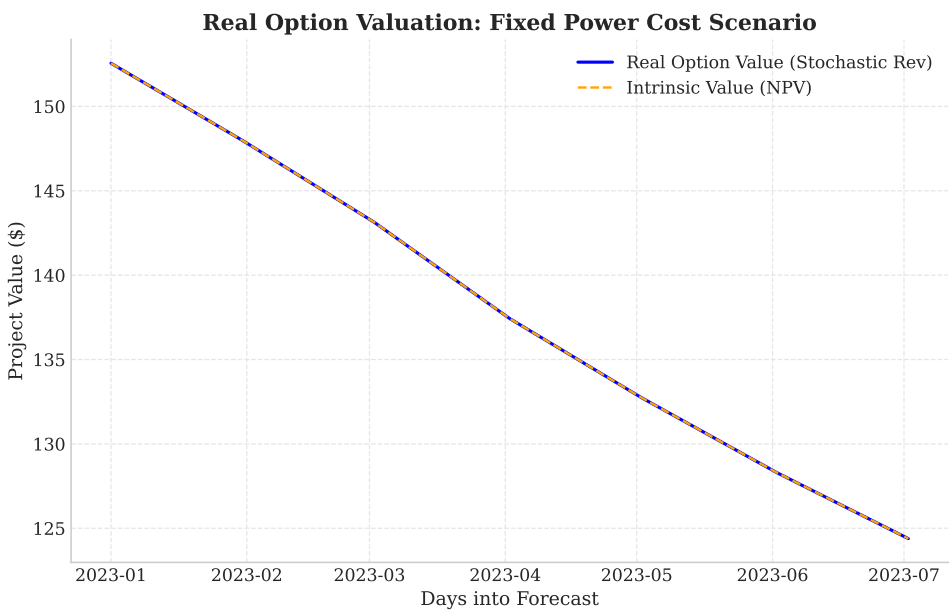

Figure 7: Evolution of the Real Option Value versus Intrinsic Value under full stochasticity. The Spark Spread curtailment option protects the firm against extreme upside spikes in power costs.

When forced to operate continuously without the ability to shut down (Intrinsic NPV), the project's value is estimated at **$153**. However, when the HMC algorithm properly

values the Spark Spread Option under full stochastic conditions, the Real Option value is **$153**. This generates a flexibility premium of **$0**.

Decomposition of Option Value: Comparing the three scenarios reveals the true sources of strategic value in cryptocurrency mining. Traditional DCF and static NPV misprice these assets because they fail to account for management's ability to asymmetrically capture upside volatility while truncating downside risk. By securing the operational flexibility to suspend operations during adverse price movements—in both the crypto (Scenario 2) and energy markets (Scenario 3)—miners can extract optimal value from the inherent volatility of the asset class.

7. Conclusion

This paper presents a framework for valuing investments in Bitcoin mining facilities using real options analysis. By employing the Hedged Monte Carlo method, we address the complexities arising from stochastic Bitcoin prices and electricity costs, the option to defer investment, and the operational flexibility inherent in mining operations. The HMC approach provides a robust and variance-reduced method for determining the optimal investment timing under uncertainty.

A. Appendix: Overview of Bitcoin Mining

Bitcoin mining is the decentralized consensus mechanism by which transactions are validated and added to the blockchain, and new bitcoins are issued. It relies on the **Proof-of-Work (PoW)** protocol, requiring participants (miners) to expend significant computational effort to solve complex cryptographic puzzles [6].

A.1. Process and Rewards

Miners aggregate transactions into a candidate block and repeatedly hash the block header until they find a value below a network-defined target (the difficulty). This requires specialized hardware, predominantly **Application-Specific Integrated Circuits (ASICs)**.

The successful miner broadcasts the block, and upon validation by the network, receives a block reward. This reward consists of the block subsidy (newly minted bitcoins) and transaction fees [5, 1]. The network difficulty adjusts approximately every two weeks to maintain an average block discovery time of ten minutes. The block subsidy halves roughly every four years; as of the 2024 halving, the subsidy is 3.125 BTC.

A.2. Mining Strategies and Time to Reward

The time until a mining farm generates revenue depends on its strategy:

- **Pool Mining:** Miners combine their hash rate in a pool and share rewards proportionally. This reduces the variance of returns, allowing for consistent, frequent payouts (hours or days).

- **Solo Mining:** The farm keeps the entire block reward if it successfully mines a block. However, the time to discover a block is stochastic and can be extremely long, following an exponential distribution.

A.3. Illustrative Numerical Example

This example calculates the expected revenue for a hypothetical mining farm.

Assumptions (Illustrative):

- Global Network Hash Rate (H_{Global}): 965 EH/s (1 EH/s = 10^{18} H/s).
- Block Reward (R): 3.125 BTC .
- Average Blocks per Day (B_{Day}): 144 .
- Farm Hash Rate (H_{Farm}): 20 PH/s (0.020 EH/s).

Calculations: The farm's share of the global hash rate (α):

$$\alpha = \frac{H_{Farm}}{H_{Global}} = \frac{20 \text{ PH/s}}{965{,}000 \text{ PH/s}} \approx 2.07 \times 10^{-5} \quad (0.00207\%).$$

Expected blocks mined per day by the farm ($E[B_{Farm}]$):

$$E[B_{Farm}] = B_{Day} \times \alpha = 144 \times 2.07 \times 10^{-5} \approx 0.00298 \text{ blocks/day}.$$

Expected time to mine one block (solo mining, mean of exponential distribution):

$$\frac{1}{E[B_{Farm}]} \approx \frac{1}{0.00298} \approx 335 \text{ days}.$$

Median time-to-first block (solo mining):

$$\frac{\ln(2)}{E[B_{Farm}]} \approx \frac{0.693}{0.00298} \approx 232 \text{ days}.$$

Expected daily revenue in BTC (assuming pool mining and ignoring fees):

$$E[\text{Revenue}_{BTC}] = E[B_{Farm}] \times R = 0.00298 \times 3.125 \approx 0.00933 \text{ BTC/day}.$$

Interpretation: For this hypothetical farm, solo mining presents high variance, with an average wait of 335 days to find a block. Pool mining provides a much more stable expected revenue stream.

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